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PALEOMAGNETIC INVESTIGATIONS AND K–AR DATING ON THE VARISCAN PLUTONIC MASSIF OF THE CHAMP DU FEU AND ITS VOLCANIC-SEDIMENTARY ENVIRONMENT, NORTHERN VOSGES, FRANCE

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ABSTRACT

Edel, J.B., Montigny, R., Royer, J.Y., Thuizat, R. and Trolard, F., 1986. Paleomagnetic investigations and K–Ar dating on the Variscan plutonic massif of the Champ du Feu and its volcanic-sedimentary environment, northern Vosges, France. *Tectonophysics*, 122: 165–185.

Paleomagnetic investigations have been carried out on samples from the plutonic series of the Champ du Feu Massif and the Devonian–Ordovician host rocks. In order to date the different phases of magnetization, a series of samples has been selected for K–Ar analysis on micas, amphiboles and feldspars. The diorites, granodiorites, diabases and hornfelses are characterized by high susceptibilities of up to 10^{-1} (SI). Thermal demagnetization reveals several groups of directions of characteristic magnetization. The A direction: $D = 202°$, $I = -23°$ (49°N, 153°E) is consistent with the Permian virtual geomagnetic pole for Europe. The B_1 direction: $D = 59°$, $I = -22°$(11°N, 128.5°E) was obtained from the effusive parts of the Kagenfels granite dated at 284 Ma. Diorites, microdiorites and volcanics, emplaced a short time after the intrusion of the Natzwiller granite, dated at 344 and 351 Ma, display three groups of directions; the C_1 direction: $D = 328°$, $I = -15°$ (27°N, 223.5°E), the C_2: $D = 323°$, $I = -38°$ (13°S, 222°E) and the C_3: $D = 272°$, $I = -41°$ (16°S, 260°E). The C_1 and C_2 directions agree well with those obtained previously for Visean volcanics from the southern Vosges. In the host rocks we have identified B and C overprintings. Despite the lack of tectonic control, the paleomagnetic data demonstrate at least two important motions of the Vosges: first a rotation of undetermined direction relative to "stable Europe", which affected the Variscan belt between the late Visean and 320 Ma, and second a clockwise rotation which occurred during the Stephanian–Autunian in late Variscan time.

INTRODUCTION

The present study is part of a systematic paleomagnetic investigation on the Variscan belt of Western Europe which aims at discovering possible deformations and displacements induced by the relative motions of Laurasia and Gondwana in Carboniferous times. A brief review of published paleomagnetic results leads us to adopt a more mobilistic view than is commonly accepted of the Hercynian history

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for that period. Edel et al. (1981a) proposed large rotations of the Corso-Sardinian basement relative to "stable Europe" for the time span 310 to 290 Ma. The Iberian arc, in northern Spain, results from a counterclockwise rotation of the Iberian Hercynian structures (Ries et al., 1980; Perroud and Bonhommet, 1981): the final tightening of the arc probably occurred during the late Carboniferous. In the northern edge of the Massif Central (France), amphibolites, plutons and late Visean volcanics yield declinations ranging from 240° to 290° which probably correspond to Namurian–Westphalian overprintings: they lend consistent support to a 60°-clockwise rotation of the northern part of the Massif Central in Westphalian–Stephanian times (Edel et al., 1981b; Edel, 1984). Volcanics from the Laval basin (Armorican Massif) give a similar result but with a smaller rotation (Edel and Coulon, 1984).

In the southern Vosges, a mean direction of $D = 323°$, $I = -17°$ with a significant dispersion in inclination, has been obtained, suggestive of a 60°-counterclockwise rotation of the Vosges relative to northern Europe, which occurred during the late Visean–late Westphalian time span (Edel et al., 1984). To the northeast of the Variscan chain, sedimentary layers from the Harz mountains indicate a 20°–30°-counterclockwise rotation, probably of Carboniferous age (Bachtadse et al., 1983).

The results presented here were obtained from plutonic and volcano-sedimentary units from the northern Vosges. The objectives of the study were first to improve our knowledge of the magnetic properties of the plutons, second to assess the impact, in terms of rock magnetism, of their intrusion on the surrounding host rocks and third to check whether the basement on both sides of the Lalaye–Lubine fault, which delimits the northern Vosges from the middle and southern, underwent similar motions in Carboniferous times. K–Ar dating of plutonics on extracted minerals was carried out in order to assign ages to the successive magnetizations.

GEOLOGICAL SETTING OF THE NORTHERN VOSGES

The Vosges Massif is divided structurally into two parts by an important fault trending E–W, called the "Lalaye–Lubine dislocation". The domain, situated to the north, belongs to the Saxothuringian zone (Kossmat, 1927) whereas the southern one is part of the Moldanubian zone. This zone of dislocation successively acted as a south-dipping thrust fault and then as a dextral strike-slip fault with an offset of about 100 km (Edel, 1982). From the Lalaye–Lubine fracture to the north, the following geological units can be followed (Fig. 1):

(1) *The schists of Villé and Steige*. The epizonal schists of Villé, considered to be of Brioverian age are limited to the north by a south dipping thrust fault which separates them from the Silurian–Ordovician schists of Steige. The latter mostly consist of violet to red phyllites of pelitic or greywacke composition. To the north they have been metamorphosed by the granodiorite of the Champ du Feu Massif (Rosenbusch, 1964; Von Eller et al., 1970). This phenomenon is characterized by the

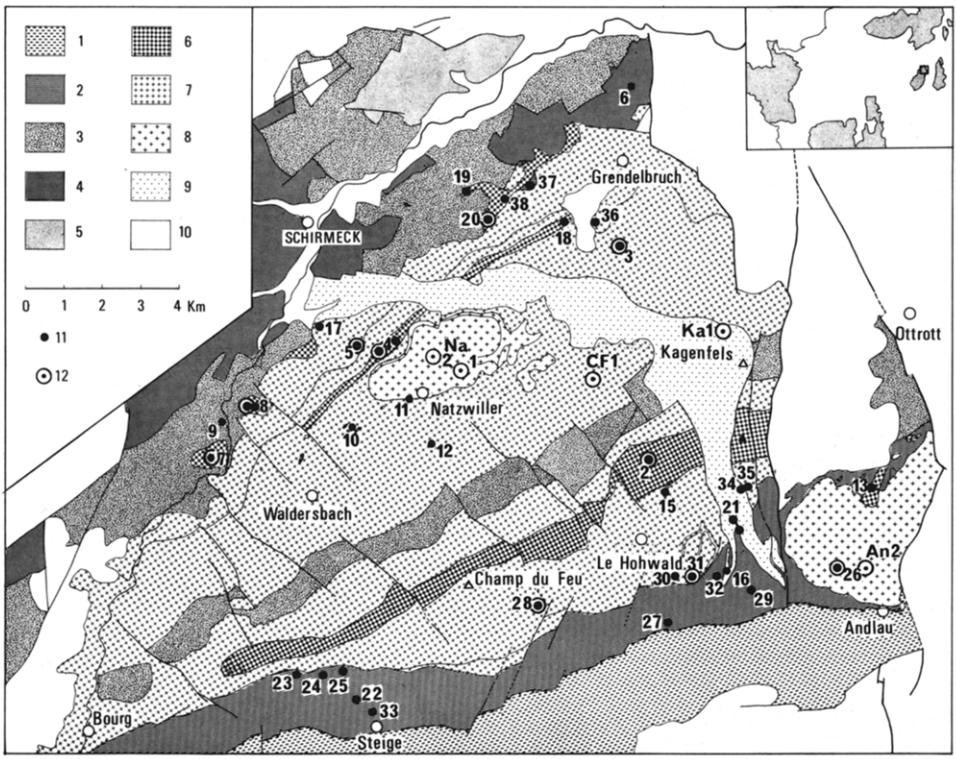


Fig. 1. Geological sketch map of the Champ du Feu Massif (northern Vosges) and site location. *1* = Brioverian schists of Villé; *2* = Ordovician–Silurian schists of Steige; *3* = Devonian; *4* = Devonian volcanics; *5* = Visean; *6* = diorites and microdiorites; *7* = granites and granodiorites; *8* = intrusive granites of Natzwiller and Andlau; *9* = late Variscan intrusive granite of Kagenfels; *10* = post-Carboniferous series; *11* = paleomagnetic sites; *12* = K–Ar dating sites.

reduction of hematite into magnetite, bringing about anomalies of the magnetic field (Lauer and Taktak, 1971; Edel and Lauer, 1974; Lauer, 1975). Rb–Sr dates for the schists average 440 Ma, and are viewed as marking the end of diagenesis (Clauer and Bonhomme, 1970). A second thermal overprinting at 350 Ma would correspond to the end of the regional metamorphism.

(2) *The diorites, granodiorites and granites of the Champ du Feu Massif*

The Champ du Feu Massif displays a series of dioritic and granodioritic bands that stretch WSW–ENE. They have been intruded successively by the granodiorites of Andlau and Natzwiller and by the acidic dyke of Kagenfels. Von Eller et al. (1970) drew the following sketch for emplacement of the plutonic complex. At first, diorites intruded volcano-sedimentary units of Devonian and Silurian age. To the north of the present volcano-sedimentary series which separates the massif into two parts, this first magmatic episode was succeeded by the emplacement of various granitic bodies such as the granites of Champ du Feu Nord, Waldersbach and

TABLE 1

K–Ar analysis and ages

Site—sample	Geological unit	System *	K_2O (wt.%)	$^{40}Ar.rad$ (10^{11} mol.g)	$\frac{^{40}Ar.rad}{^{40}Ar.tot.}$	Age (Ma)
Ka 1	Kagenfels granite	biotite	8.39	371.8	94	284 ± 9
26 (81.001)	Andlau granodiorite	biotite slightly chloritized	5.40	272.3	96	320 ± 10
An 2	Andlau granodiorite	biotite chloritized (m)	2.53	139.99	77	348 ± 12
		biotite chloritized (n.m)	4.68	258.65	81	348 ± 12
28	Hohwald granodiorite	biotite chloritized	3.49	185.79	72	336 ± 11
		amphibole ($d = 3.15$)	0.500	24.69	45	314 ± 15
31 (85.005)	Hohwald granodiorite	feldspar	1.91	84.49	90	284 ± 9
		chloritized biotite	0.662	30.81	80	297 ± 10
		amphibole (m.)	0.412	20.23	68	312 ± 13
		amphibole (n.m.)	0.380	19.00	85	318 ± 15

20	Diorite	biotite + amphibole	2.32	119.43	77	326 ± 11
		Biotite + chlorite (d = 2.89)	4.59	250.94	88	345 ± 11
2 (81.006)	Diorite	amphibole (d = 3.20)	0.741	38.76	95	331 ± 11
3 (81.009)	"Nadeldiorite" inclusion	amphibole (d = 3.13)	1.34	69.07	97	327 ± 10
3 (81.010)	Waldersbach granite	biotite (d = 3.0)	6.16	310.8	98	320 ± 10
CF1 (81.008)	Champ du Feu N. granite	chloritized biotite $2.89 < d < 3.10$	5.90	290.4	87	313 ± 10
8b	Diorite	weath. hornblende	0.417	18.47	21	284 ± 18
7a	Diorite	hornblende $2.96 < d < 3.03$	0.175	7.684	22	282 ± 23
5	Granit. dike	amphiboles + chlor. biotites (d = 2.86)	3.22	163.0	86	321 ± 10
4	Microgranite	chlor. Amphibole + chlor. biotite	3.52	183.5	84	328 ± 11
Na 1	Natzwiller granite	Slightly chloritized biotite	6.17	336.3	87	344 ± 11
Na 2	Natzwiller granite	biotite	8.07	450.8	98	351 ± 12

* n.m.—non magnetic fraction (relative); d—density.

Grendelbruch. To the south and probably at the same time several granodioritic plutons were implaced, namely: the Champ du Feu Sud, Hohwald and Andlau granodiorites. The later ones metamorphosed the schists of Steige. Contemporaneously or subsequently thereafter, the granodiorite of Natzwiller intruded the northern part of the massif. Later, probably in the Westphalian–Stephanian time, the granite of Kagenfels crossed over the preceding formations. This intrusion gave rise to hypovolcanic rocks of rhyolitic type along its southern edge.

(3) *The Devonian–Dinantian of the Bruche valley.* The volcano-sedimentary series of the Bruche valley consists of a Givetian spilite–keratophyre association and Givetian to Visean schists and greywackes. The whole series underwent a thermal metamorphism attributed to granitic intrusion. Rb–Sr dates of Visean and Devonian schists yield an age of 342 Ma (Bonhomme and Prévôt, 1968).

Paleomagnetic samples have been obtained from eight sites of diorites (2, 7, 8, 13, 14, 19, 20, 38), seven sites of microdioritic inclusions from the Waldersbach granite (3, 4, 10, 11, 12, 18, 36), three sites of diorites which are xenolithes in the Hohwald and Andlau granodiorites (26, 28, 31), four sites from acidic dikes (5, 15, 17, 23), three sites from the Kagenfels rhyolite (16, 21, 34), three sites of Devonian volcanics (6, 9, 37), one site of lamprophyre (30) and seven sites from the Steige schists (1, 22, 24, 25, 27, 29, 32, 33).

ANALYTICAL TECHNIQUES OF K–Ar DATING

Unless otherwise stated, the 100–160 μm sieve fraction of the ground rock samples was selected for magnetic separation of micas, amphiboles and feldspars. Whenever necessary, further refinement was done with heavy liquids. Our analytical procedures were described in details elsewhere (Westphal et al., 1979). Briefly, potassium was measured by flame photometry with lithium internal standard (Besnus, 1978). Argon was extracted in a bakeable glass vacuum apparatus and determined by isotope dilution technique (using ^{38}Ar as a tracer) with a MS 10S mass spectrometer. All samples were treated by the static mode.

The results are listed in Table 1. Plus-minus figures represent estimates of analytical precision at one standard deviation. They were calculated following the procedure given in Cox and Dalrymple (1967). Dates were obtained with the set of constants recommended by Steiger and Jaeger (1977).

K–Ar DATES AND DISCUSSION

The isotopic mineral ages of plutonic rocks can be interpreted either as ages of intrusion or of cooling. Harrison and MacDougall (1980) have shown that K–Ar dates yielded by amphiboles in granitic plutons are indistinguishable from those of crystallization. On the other hand, biotite ages differ by 3 to 4 Ma from intrusion ages. Consequently, the moderate to tiny size of the Vosges plutons and the order of

magnitude of absolute errors, lead us to assume that in many cases the calculated ages represent both ages of cooling and intrusion.

In the northern Vosges the ages range from 351 ± 12 Ma to 282 ± 23 Ma. Either biotites or amphiboles have undergone secondary transformations subsequent to crystallization (Table 1). The interpretation of the ages is, therefore, somewhat ambiguous. Four of them, however, can be regarded as essentially undisturbed, taking into account both freshness and geological location. The two biotites from the Natzwiller granite are either fresh (NA 2) or slightly chloritized (NA 1) as can be seen from the potassium content. Both give similar ages within the error bars, 344 ± 11 Ma and 351 ± 12 Ma. This concordancy and the small size of the pluton compel to interpret them as reflecting the time of intrusion. A fresh biotite from the Kagenfels granite indicates an age of 284 ± 9 Ma which may be representative of the crystallization. In the same way, the similarity of the calculated ages given by two fractions of a biotite from the Andlau granite with highly different K contents favours to take into consideration the value of 348 ± 12 Ma. If the above interpretations are correct, the calculated ages within the 340–290 Ma range should be viewed as resulting from argon loss and consequently without any geological meaning since the Natzwiller granite is the latest of the Massif before the Kagenfels intrusion. In summary, the main phase of granitization is at least as old as 350–340 Ma.

PALEOMAGNETIC SAMPLING AND LABORATORY TECHNIQUES

Oriented samples were drilled in situ or collected as handsamples. Whenever possible, they were oriented both with a solar and a magnetic compass. The handsamples were cored in the laboratory and the cores subsequently cut into standard specimens 2.5 cm long. Remanent magnetization measurements were performed on a computerized Digico magnetometer, with noise level reduced to $3 \cdot 10^{-5}$ Am^{-1}. Susceptibilities were determined with a Digico susceptibility bridge. The alternating field and thermal demagnetizations were done with laboratory-made apparatus.

NATURAL REMANENT MAGNETIZATIONS AND SUSCEPTIBILITIES

High magnetic anomalies of up to 300 nT at 1200 m above sea level overlie the northern Vosges. They belong to a band of anomalies which extends from the Paris Basin to the Thüringerwald (Edel, 1982). Their origin is attributed to the magnetic properties of granodiorites, diorites and rocks of contact metamorphism (Lauer and Taktak, 1971; Edel and Lauer, 1974; Lauer, 1975; Gançarski, 1977).

Susceptibility measurements were systematically performed on at least one specimen per sample. High values of 10^{-2} to 10^{-1} (SI) characterize most of the diorites, microdiorites and schists affected by contact metamorphism (Fig. 2). For these rocks, the natural remanent magnetization intensities scatter from 10^{-1} to 1 Am^{-1}.

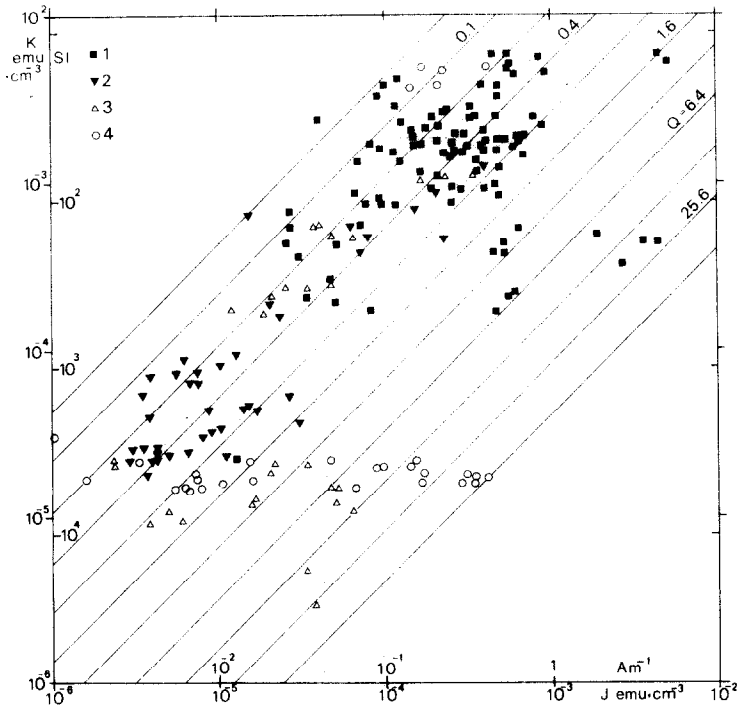


Fig. 2. Natural remanent magnetization (J), susceptibility (K) and Koenigsberger ratio ($Q = J/KF$) of rocks from the northern Vosges. *1* = diorites, microdiorites, granodiorites; *2* = volcanics and basic dykes; *3* = granites and acid dykes; *4* = schists.

Acid and basic dykes, as well as some microdiorites and volcanics form a well-defined linear array in the K versus J diagram ($Q = J/KH$) (Fig. 2), the corresponding Koenigsberger ratio being near 0.3. To the opposite, the Steige schists and the Kagenfels rhyolite display susceptibility values that cluster around $2.5 \cdot 10^{-4}$ while the NRM intensity ranges from $2 \cdot 10^{-1}$ to $4 \cdot 10^{-1}$ Am^{-1}. As a consequence, the Koenigsberger ratio reaches, in this case, values beyond 10. Few dioritic samples can be classified with this latter group (sites 8, 13, 38). It must be kept in mind, however, that the remanent magnetization intensity is, in most cases, weaker than the induced ($Q < 1$).

REMANENT MAGNETIZATIONS

In order to detect possible superimposed secondary components of magnetization, detailed stepwise thermal and alternating field demagnetization were undertaken on all samples.

After elimination of a soft viscous component, both types of demagnetization indicate a harder component with a direction close to the present magnetic field and

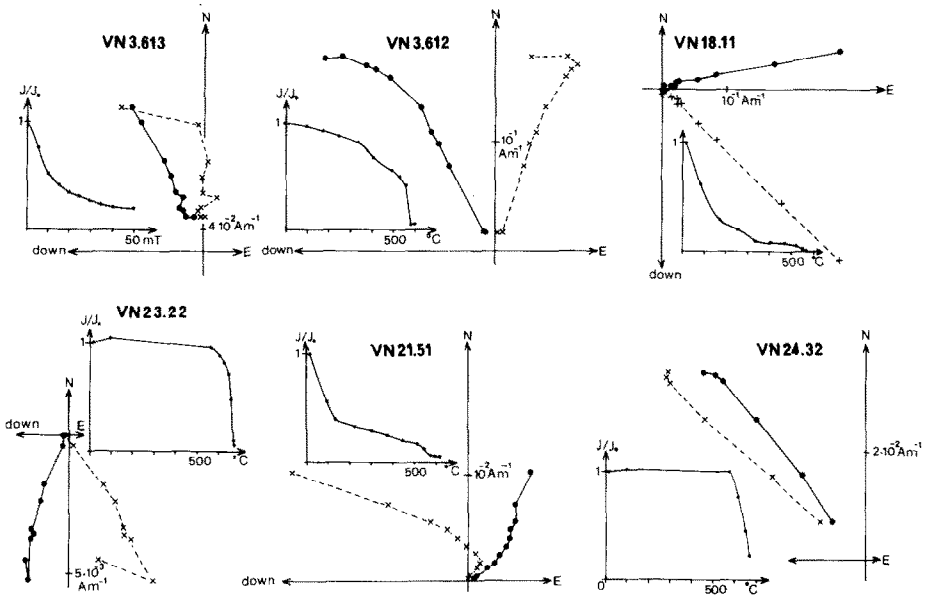


Fig. 3. Typical demagnetizing curves of microdiorities (VN 3.613); VN 3.612; VN 18.11), of acid dykes (VN 23.22; VN 21.51) and of a schist (VN 24.32). Orthogonal projections on the horizontal plane (dots) and on the vertical NS plane (×) or EW plane (+).

unblocking temperatures of 300°–500°C (sites 6, 7, 9, 17, 21, 34, 35) (Fig. 3, specimen 21.51). In this case, AF demagnetization does not restore all components of the NRM. Except of few sites of coarse-grained diorites, where the harder component is absent, AF demagnetization was either incomplete or insufficiently discriminating. In site 3, for instance, AF treatment on specimen VN 3.613 does not restore the characteristic negative inclination yielded by thermal treatment of specimen VN 3.612 from the same sample (Fig. 3).

Some sites of diorites and microdiorites as well as inclusions of the Hohwald and Andlau granodiorites, exhibit components with low coercivities (15–20 mT) and low unblocking temperatures (300°–400°C). In the southern part of the massif, the dispersion of directions precluded the computation of a mean characteristic direction (sites 2, 14, 26, 28). On the other hand, sites 8 and 18 located in the north, display a nice cluster, where the magnetization is colinear with that carried by magnetite (Fig. 3, specimen VN 18.11). These low-temperature magnetizations, probably are due to a hydrothermal alteration process, which also may have caused argon loss in sites 2, 26, 28 and 8b.

Magnetite is the main carrier of the characteristic remanent magnetization in diorites, microdiorites and basic volcanics (Fig. 3, VN 3.612). For rhyolites and acid dykes thermal demagnetizations reveal either magnetite together with hematite (Fig. 4, VN 21.51) or only hematite (Fig. 4, VN 23.22).

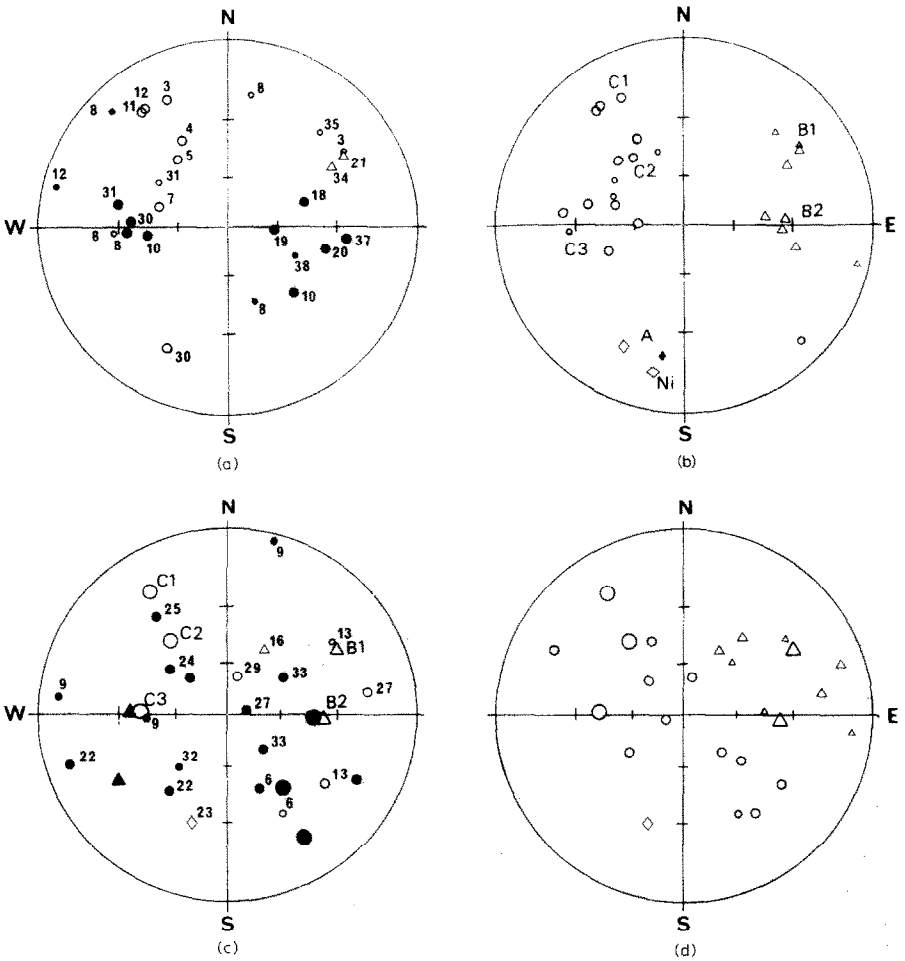


Fig. 4. Stereographic projection of the "in situ" site mean directions. a. In the batholite. b. In the batholite after reversing in the upper hemisphere. c. In the hostrocks. d. In the hostrocks after reversing in the upper hemisphere. The small symbols represent directions with $n < 3$, $N = 1$ or $\alpha_{95} > 20$. The great symbols represent the mean C_1, C_2, C_3, B_1, B_2 and A normal and reversed directions. *Ni* is a result from the Nideck–Donon Permian volcanics situated just at the North of the sketch map (Roche et al., 1962). Open symbols are plotted in the upper hemisphere, full symbols in the lower.

For the metamorphosed schists of site 29 a very steep drop of the magnetization is observed up to 350°C, followed by a great unstability above. On the other hand, for low-susceptibility schists within occur titano-hematites with unblocking temperatures between 600° and 680°C, the magnetization is extremely stable. For sites 27 and 33 the thermal demagnetization indicates several high temperature components which can hardly be separated.

Two sites from dioritic inclusions (26, 28), two from the Champ du Feu diorite (2,

14), one from hornfelses (1) and one from an acidic dyke (15) failed to provide any characteristic magnetization. For the last site, the NRM is too weak. The others are characterized by high susceptibilities, a low Koenigsberger ratio and occurrences of hydrothermalized minerals. All remaining sites are assumed to display characteristic magnetizations which can be related either to emplacement or to subsequent tectonic and/or thermal phases.

DIRECTIONS OF REMANENT MAGNETIZATIONS (Table 2)

Results from sites located within the massif will be examined separately from those obtained from Devonian and Ordovician–Silurian volcanic and sedimentary host rocks, which were folded, metamorphosed and tilted by either Variscan or Alpine orogenic phases.

Champ du Feu massif (Fig. 4a, 4b)

In the following presentation, the whole massif is supposed to be a uniform undeformed block. Several sites (Nos. 6, 7, 9, 17, 21, 34, 35), mostly situated in the vicinity of N–S faults related to the Rhinegraben taphrogenesis display overprintings with unblocking temperatures of up to 520°C and rather high coercivities. The directions, close to the present magnetic field, are suggestive of recent overprintings and will not be, therefore, taken into account.

Mean site directions, revealed by high-temperature magnetizations, are very different from those expected from the Carboniferous mean poles of northern Europe. This feature is, however, not displayed at sites 8 and 30. A high temperature A-direction close to that given by the Permian Nideck–Donon ignimbrites (Roche et al., 1962) is recovered from site 30, along a B-westerly direction of a middle temperature component. The A-direction may be due to the intrusion of an acidic dyke, related to the latest magmatic phase. It has to be noted in this respect, that site 23, corresponding to a dyke intruded into the Steige schists (Fig. 4c), exhibits a mean high-temperature direction similar to that of site 30.

Figure 4b, depicts all mean directions of Fig. 4a reversed in the upper hemisphere. Beside the A-direction, this representation points to two other groups of directions: the B-directions characterized by declinations around N60° (B_1) and N90° (B_2), and the C-directions with declinations around N325° (C_1 and C_2) and N270° (C_3).

The B_1-direction is found in the Kagenfels rhyolite and corresponds to the latest granitic phase of the northern Vosges, as indicated by the biotite age of 284 ± 9 Ma. The mean direction, recovered from sites 21 and 34 is $D = 59°$, $I = -22°$ (VGP: 11°N, 128.5°E). Similar directions are recorded in the vicinity of the Kagenfels granite (sites 3 and 35), and indicate a magnetic overprint of the granite onto the surrounding terrains. After inclusion of the results yielded by those two sites, the mean B_1-direction becomes $D = 55°$, $I = -21°$ (VGP: 13°N, 131°E).

TABLE 2

Paleomagnetic site mean results after demagnetizing *

Site	Formation	*Tb/Hc*	N/N_0	*n*	*D*	*I*	*k*	α_{95}	pol.	VGP lat., long.
3	microdiorite	580	9/10	15	334	−15	94	4		
	microdiorite + granite	350–400	2/10	3	57	−18	32	22		
11	microdiorite	500	2/2	5	324	−15	76	9		
12	microdiorite	570	3/4	7	325	−15	12	18		
	Mean C_1(3 + 11 + 12)			3	328	−15	232	8		27°N, 223°5'E
4	microdiorite	580	5/6	7	332	−35	110	6		
5	acid dyke (Natzwiller)	580	5/5	9	323	−42	58	7		
		450	5/5	12	335	70.5	11	14		
10	microdiorite	580	3/8	6	135	37	81	7		
	microdiorite + granite	350–560	4/8	7	263	44	39	10		
	Mean C_2(4 + 5 + 10)			3	323	−38	111	12		13°N, 222°E
	Mean $C_1 + C_2$ (3 + 11 + 12 + 4 + 4 + 10)			6	326	−27	33	12		20°N, 223°E
7a	diorite				–	–				
7b	basic dyke + diorite	580	6/7	10	282	−48	18	12	N + I	
8a	diorite	560–580	3/3	3	266	−28	20	28		
		300–350	2/3	3	160	44	35	21		
8b	diorite	350, 580	4/6	8	266	34	51	8	N + I	
		350, 570	1/6	2	10	−19				
		580	1/6	2	315	8				
18	microdiorite + granite	350, 580	6/6	10	72	44	46	7		
19	diorite or volcanic	580	4/4	8	93	62	31	10		
20	diorite	580	6/6	9	103	34	38	8		
37	spilite	500–520	7/7	10	48	43	21	11		
		580	7/7	9	96	25	46	8		
	Mean C_3 (7b + 8a + 18 + 19 + 20 + 37)			6	92	41	24	14		16°S, 260°E
30	lamprophyre + granite	350–400	3/7	5	10	62	61	10		
		500–530	4/7	6	273	36	31	12		
		530	4/7	7	205	−18	83	7		
31	granodiorite +	300–400	6/7	9	286	29	15	14		
	inclusion	350	1/7	2	303	−43				
35	granodiorite	350	2/4	5	341	56	34	13		
		580–600	1/4	2	44	−20				
36	granite	300–370	3/6	5	64	60	42	12		
		15 mT	3/6	3	52	17	102	12		
38	diorite	580	1/6	2	112	47				
13	diorite		2/6	4	125	−25	243	6		
			2/6	4	117	15	81	10		
			1/6	2	53	−22				

TABLE 2 (continued)

Site	Formation	*Tb/Hc*	N/N_0	*n*	*D*	*I*	*k*	α_{95}	pol.	VGP lat., long.
21	rhyolite Kagenfels	650	7/8	10	58	−19	24	10		
34	rhyolite Kagenfels	500–520	7/7	9	347	58	91	5		
		600	4/7	6	60	−25	136	6		
	Mean B_1 (21 + 34)			2	59	−22				11°N, 128°5′E
	Mean B_1, (21 + 34 + 3 + 35)			4	55	−21	118	8.5		13°N, 131°E
	Mean B_2 (8b + 10 + 30 + 31)			4	272	36	60	12		7°S, 269°E
6	spilites, Devonian	580	2/5	5	157	44	77	9		
			2/5	2	151	−28				
9	spilites, Devonian	580	3/5	3	1	56	197	9		
		580	1/5	2	15	2				
		350	1/5	3	267	43	172	9		
		580	1/5	3	276	6	25	25		
16	rhyolite	580, 670	3/3	4	29	−46	142	8		
23	acid dyke	675	2/2	4	198	−28	341	5		
22	schists Steige	660	2/2	5	217	36	54	4		
		680	2/2	6	252	6	58	9		
24	schists Steige	670	4/4	8	308	47	75	6		
		675?	4/4	6	312	59	119	6		
25	schists Steige	675	4/4	7	324	24	88	6		
27	schists Steige	650	5/5	4	81	78	140	4		
		675	5/5	6	81	−16	60	9		
29	schists Steige	350–400	5/5	7	14	−66	83	7		
32	schists Steige	670	1/1	2	222	49				
33	schists Steige	670	2/3	4	134	60	104	9		
		670	2/3	4	56	51	81	10		
	Mean A (23 + 30)			2	202	−23				49°N, 153°E

* *Tb/Hc*—unblocking temperature or coercitive field; N/N_0—number of samples taken into account/total number of samples; *n* + number of specimen taken into account; *D*, *I*—in situ declination and inclination; *K*, α_{95}—Fisher statistic parameters.

The B_2-directions are reversed and can be observed in granite, granodiorite (sites 10 and 31), diorite (site 8b) and in lamprophyre of site 30. Most of the samples show evidence of water-rock interactions. The site 8b is situated close to the northern edge of the massif and the hornblende yields a K–Ar age of 284 Ma similar to that of the

Kagenfels granite. Sites 30 and 31 are located in the Hohwald granite which also displays young ages (284–318 Ma). The B_2-magnetizations are unblocked in the range 350–560°C. We suggest that they resulted from the same hydrothermal processes which caused argon loss. The mean B_2-direction is $D = 272°$, $I = 36°$ for four sites (VGP: 7.5°N, 269°E).

The *C*-directions are found exclusively in the northern part of the massif. As revealed by thermal demagnetization, magnetite is the carrier of those magnetizations. Inside of the *C*-group, three subgroups can be distinguished, which represent well-delimited zones of the massif. The C_1-direction, $D = 328°$, $I = -15°$ (sites 3, 11, 12) (VGP: 27°N, 223.5°E), corresponds to outcrops located in the internal part of the Waldersbach granite. A steeper inclination characterizes the C_2-direction, $D = 323°$, $I = -38°$ (sites 4, 5, 10) (VGP: 13°N, 222°E) which is recovered for sites closer to the northern edge. The third group, with a mean C_3-direction, $D = 272°$, $I = -41°$ (VGP: 16°S, 260°E) represents sites 7, 8a, 18, 19, 20 and 37 all situated along the border with the Devonian of the Bruche valley. The distinction between $C1$- and C_2-groups may be artificial, however. If so, the mean C_1- + C_2-direction $D = 326°$, $I = -27°$ (VGP: 20°N, 223°E) probably is related to the Natzwiller intrusion, since these sites are located in its vicinity or on its axis (site 3).

No correlation exists between the spatial distribution of magnetizations and that of K–Ar ages. For the diorite of site 20, located on the edge of the massif and characterized by a typical C_3-magnetization, ages of 326 and 345 Ma were found. The older date is consistent with the 342 Ma value obtained by Rb–Sr dating for the metamorphism that affected the Devonian–Dinantian of the Bruche valley (Bonhomme and Prévôt, 1968). Sites from the vicinity of the Natzwiller granite (4, 5) yield ages of 321 and 328 Ma, and C_2-directions while the site 3 indicates a C_1-direction along with a 327 Ma age. Those younger dates suggest partial argon losses probably induced by hydrothermal alteration.

Since all of the *C*-magnetizations are carried by magnetite, one can infer that they originated during a cooling process rather than a low-temperature alteration phenomenon. If this assumption is correct, one can conclude that the *C*-magnetizations occurred within a short time span after the intrusion of the Natzwiller granite (345 Ma).

Volcanic-sedimentary host rocks (Fig. 4b, 4c)

Two sites (Nos. 6 and 9) within the Devonian volcanics of the Bruche valley were sampled whereas the other samples are from the Ordovician–Silurian schists of Steige (sites 1, 22, 24, 25, 27, 29, 32, 33) and the related magmatic units intrusive into the schists: a porphyric acid dyke (site 23), an outcrop of the Kagenfels rhyolite (site 16) and a dioritic dyke (site 13). The characteristic "in situ" directions are much more scattered than inside of the Champ du Feu massif.

Site 23 discloses a typical Kiaman or *A*-direction, very close to that obtained by

Roche et al. (1962) on the Nideck–Donon ignimbrite, which borders the Bruche syncline to the north. Direction of site 16, which belongs to the Kagenfels magmatism, deviates from those of sites 21 and 34. It can be brought into coincidence by tilting the direction along the horizontal axis of the dyke from 40° to 50°. Thus, one can infer that deformation of the schists was still active after emplacement of the Kagenfels granite. Several sites located in the proximity of this intrusion yield normal (sites 27, 13, possibly 29) and reversed (sites 9, 22, 32) scattered *B*-directions. All of them correspond to post-folding overprints probably induced by the Kagenfels magmatism.

To the east, away from the Kagenfels granite and close to the contact with the Hohwald granodiorites, sites 24 and 25 show directions with *c*-declinations and positive inclinations. Directions of sites 6 and 13 probably belong to the same group.

High-temperature magnetizations with steep inclinations are recovered from two sites (27, 33). They may correspond to C_3-components.

In light of the Rb–Sr data, Clauer and Bonhomme (1970) proposed a complex metamorphic and tectonic history for the schists, i.e., diagenesis at 440 Ma and subsequently a regional metamorphism, which produced a strong albitization, at 372 Ma. The fine fractions indicate that two mobilizations took place thereafter at 352 Ma and 276 Ma. It is very tempting to correlate magnetic overprintings with isotopic homogenizations. Thus, the latest magnetic overprint may be related to the Kagenfels magmatism and consequently to the *B*-magnetizations. Similarly, the *C*-magnetizations of sites 24 and 25 may have been induced by the intrusion of the Hohwald granodiorite, about 350 Ma. In the same manner the magnetizations with steeper inclinations (27, 33) might be of that age or older, 372–440 Ma.

DISCUSSION AND CONCLUSION

Local tectonics

The major shortcoming of the present study lies in the absence of information on original attitudes. We have assumed that the igneous Champ du Feu massif behaved in a homogeneous manner after emplacement of the granites. Only one major fault separates the eastern part, containing the Andlau granodiorite, from the rest of the massif, and it had significant activity only during the Rhinegraben formation. A series of NW–SE dextral strike-slip faults affected the massif before the Kagenfels intrusion and probably before that of Natzwiller. These may have resulted from a clockwise rotation of that part of the Variscan chain. No significant tectonic features separate zones that display C_1, C_2, C_3, B_1, B_2-magnetic components. Hence the different magnetizations might have been acquired along with the successive positions of the batholith. For instance the massif would have been tilted to the north-northwest during thrusting of the Villé schists over the Steige schists. However, we have no evidence of such tilting. In the southeastern part of the pluton, the

Steige schists of site 25 which seem to overlie the Hohwald granodiorite display subhorizontal bedding and schistosity. Directions C_1, C_2 given by the plutonic rocks and those of sites 24, 25 may fall as well on a small circle with a N80 axis corresponding to the direction of schistosity as on a great circle. This implies that either a local tilting of the massif or a northward/southward drift of the whole region can equally explain differences in inclination. Since a possible tilting of the massif does not severely affect the declinations of C_1, C_2-directions, one can safely draw definitive conclusions as to the direction of the massif when those magnetizations occurred, i.e., within the 345–325-Ma time span. Concerning the *B*-magnetization, a weak clue—i.e. a subhorizontal "flame" in the rhyolite of site 34, may indicate that no important tilting has affected the massif after intrusion of the Kagenfels granite. In the schists all magnetizations followed major episodes of folding. Unfortunately, because of the poor knowledge of the latest tectonic events no correction can be applied and consequently, no mean direction can be computed.

Late Variscan rotation

We may now attempt to interpret the significant differences observed in the directions of groups *A*, *B*, *C*. The *A*-group only is represented by two dykes , one

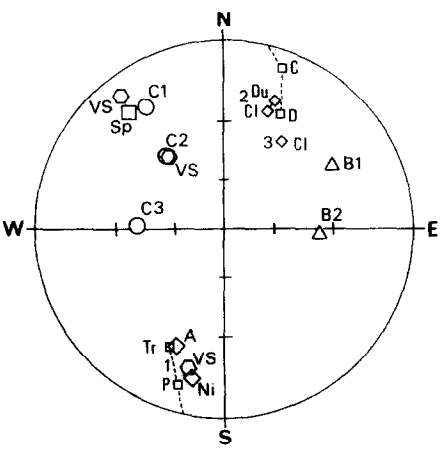


Fig. 5. Stereographic projection of the mean *A*, *B* and *C* directions obtained in the northern Vosges. *VS* and *Sp* represent directions computed for the northern Vosges, from results of the southern Vosges (Edel et al., 1984) and of the Spessart (F.R.G.) (Bosum et al., 1973). *Ni* is the mean direction of the Nideck–Donon Permian volcanics (Roche et al., 1962). Directions have been computed for the northern Vosges from the following European VGP. Open symbols are plotted in the upper hemisphere, full symbols in the lower. (*1*) Krs (1982)—Devonian: 26.65°N, 158.94°E; Carboniferous: 36.13°N, 162.71°E; Permian: 44.43°N, 165.61°E; Triassic: 49.48°N; 154.34°E. (2) French (1976)—late Devonian: 30.20°N, 161.40°E; early Carboniferous: 28.50°N, 163.60°E; (late Carboniferous: 41.30°N, 166.70°E). (3) Duff (1980)—early Carboniferous: 18°N, 153°E; (middle–late Carboniferous: 35°N, 155°E).

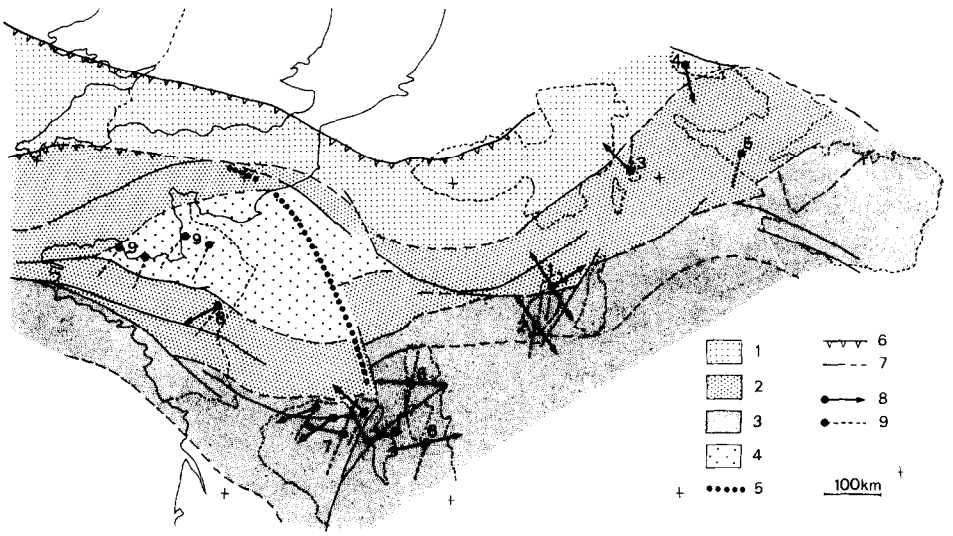


Fig. 6. Carboniferous–Permian paleomagnetic directions obtained in the western Europe Hercynides on ante-Permian series. *1*—Edel et al. (this paper); *2*—Edel et al. (1984); *3*—Bosum et al. (1973); *4*—Bachtadse et al. (1983); *5*—Kim and Soffel (1982); *6*—Edel et al. (1981b); *7*—Edel (1984); *8*—Edel and Coulon (1983); *9*—Jones et al. (1979). Key: *1* = Moldanubian zone; *2* = Saxothuringian zone; *3* = Rhenohercynian zone; *4* = Cadomian bloc; *5* = magnetic anomaly of the Paris Basin; *6* = thrustfault; *7* = major discontinuity (Edel, 1982); *8* = Hercynian magnetizations; *9* = late-Hercynian magnetizations.

located in the Hohwald granodiorite, the second in the schists, close to the granodiorite. Since the Permian Nideck–Donon ignimbrites indicate a similar direction (Roche et al., 1962), the emplacement of magnetizations can be considered to be late. The *A*-direction belongs to the Kiaman reversed interval with a corresponding VGP which is consistent with European poles, i.e. the latest Carboniferous and Permian poles (Fig. 5). It has to be noted that this VGP is unique in the northern Vosges.

The B_1-direction is exhibited as a primary magnetization in the Kagenfels rhyolite and as overprinting in the neighbouring series, granodiorites and schists. A K–Ar determination on biotite from the Kagenfels granite indicates an age of 284 Ma which corresponds to the Carboniferous–Permian boundary. Closely similar directions exist in the northern and northeastern Massif Central (Edel et al., 1981b; Edel, 1984), where they have been observed in Carboniferous volcanics, plutons and amphibolites (Fig. 6). In the latter formation, the mean direction: $D = 247°$, $I = 3.5°$ (VGP: 14°N, 110°E) represents an overprinting related to leucogranite intrusion dated at 312 Ma (Edel, 1984). Further west, on the northern edge of the Laval Basin (Massif Armorican) a post-folding overprint is observed with mean direction $D = 238°$, $I = -8°$ (VGP: 24°N, 112°E) of which the declination can be compared with the reversed B_1. This latter overprint occurred after the early

Westphalian. Wandering of the paleomagnetic directions from B to A cannot be simply explained by tilting of the Champ du Feu massif to the southeast, since the dip would exceed 70°. Thus we are compelled to assume a Stephanian–Autunian clockwise rotation of the northern Vosges relative to the "stable Europe". If the B_1-direction is considered, the angle of rotation is about 30°. It reaches 70° with the B_2-direction. Because of the uncertainty of magnetization polarity a counterclockwise rotation would also be a possible but very unlikely solution. We have interpreted the late Variscan clockwise rotations in the Armorican Massif and the Massif Central as a consequence of the latest tightening phases of the Iberian–Armorican arc.

Visean–Westphalian rotation

After K–Ar dating, the C-magnetizations probably took place in the time span 345–325 Ma, but their distribution with time remains undetermined. Nevertheless, we favour the sequence C_3–C_2–C_1, which best corresponds to the cooling process of the massif after intrusion of the Natzwiller granite. C_3 would then be the oldest component. A tectonic event, either a local tilting or a regional rotation must have occurred during the C_3–C_2-interval.

Directions very close to C_1–C_2 were obtained on Visean volcanics from the southern Vosges (Edel et al., 1984), on metamorphic rocks from the Spessart (FRG) (Bosum et al., 1973) and on volcanics and plutons from the Massif Central (Edel, 1984, figs. 5, 6). In the latter case, the magnetizations are of Carboniferous age. In the southern Vosges two mean directions, $D = 324°$, $I = -39°$ (VGP: 13°N, 221°E) and $D = 322°$, $I = -7°$ (VGP: 29°N, 231°E) were computed. The latter, which is probably the younger, essentially represents a thermal or hydrothermal overprint in relation to the magmatism of the granite des Crêtes dated at 345 Ma. One ignimbritic layer dated at 337 Ma (Montigny et al., 1984) yielded an in situ direction $D = 324°$, $I = -13°$, and a dip-corrected-direction $D = 334°$, $I = -16°$ (Hernot, 1983). In the Spessart, the NRM from three sites sampled in metamorphic series form a nice cluster around $D = 322°$, $I = -12°$ (VGP: 25°N, 231°E) (Bosum et al., 1973). These magnetizations probably took place after the latest metamorphic event, during a cooling phase between 335 Ma and 325 Ma (Kreuzer, 1975). Thus, in three different regions, various geological units such as volcanic, plutonic and metamorphic rocks display paleomagnetic directions and radiometric ages that are very similar. All of them correspond to the 345–325 Ma time span. Northern and southern Vosges, at that time, exhibited the same direction of magnetization.

The polarity of the C-components is questionable. In the sourthern Vosges the NNW directions were assumed to be normal (Edel et al., 1984), since the inclinations are not significantly different from means computed from European poles (Fig. 5). This implies a 60°-counterclockwise rotation of the basement between the Visean and the Westphalian–Stephanian. With the existence in the northern Vosges of

late-Carboniferous *B*-directions different from the European *A*-directions, the orientation and the amount of rotation have to be discussed again.

Two explanations are proposed for the wandering from C_1–C_2 to *B*-directions. Both assume a normal *B*1-direction: either a counterclockwise rotation of 90° to 130° relative to northern Europe, or a clockwise rotation of 90° to 50°. For lack of tectonic control in the northern Vosges, as well as of good Carboniferous reference data, no inference can be drawn concerning the inclinations and consequently about paleolatitudes. The rotation should have occurred after the late Visean and before the late Westphalian. From a geological point of view, such a motion is usually associated with strike-slip faulting. In the Vosges the most significant strike-slip features which were active subsequently to the 340 Ma magmatism are dextral E–W faults (the Lalaye–Lubine dislocation) and sinistral NNE–SSW faults (the fault of Ste Marie-aux-Mines in the middle Vosges and parallel faults in the Rhinegraben) (Edel, 1982). The N35 Rhinegraben fault offsets the prolongation of the Lalaye–Lubine dislocation of 35 km to the north. At 320–325 Ma, the NNE–SSW and E–W faults of the Morvan and Vosges were intruded by a series of leucogranites. If strike-slip and bloc rotations were contemporeanous, then the 320 Ma date would represent an upper limit for the motion which might be an expression of the Sudetic phase.

CONCLUSION

In the northern Vosges, two plutonic phases have been dated: the first one ended at about 345 Ma with the Natzwiller intrusion; the second, corresponding to the Kagenfels granite, took place in late Variscan time, at 284 Ma.

The paleomagnetic data imply a complex tectonic history. Except for a few magnetizations of probable Permian age, all results are in disagreement with those expected from the European virtual geomagnetic poles. These discrepancies can be explained with two general motions: the first is a rotation of undetermined direction which occurred between the late Visean and 320 Ma; the second of late Carboniferous/earliest Permian age, is a clockwise rotation of 30° to 70°, which also involved the Massif Armoricain and the Massif Central.

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